Name: ______________________

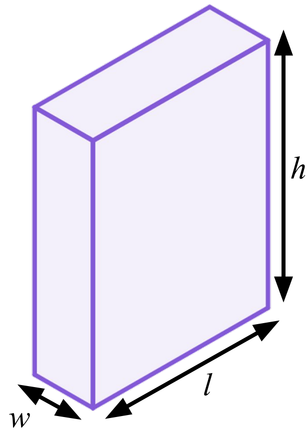
Worksheet

Evaluating expressions without changing the subject

Task A: Rearranging in order to solve

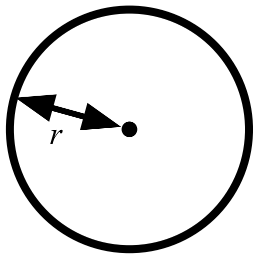
1) The formula for volume of a cuboid can be written as $V = lwh$

Rearrange the formula to make w the subject. Then evaluate to find the width of a cuboid with length 1.6 cm, height 2.25 cm, and volume 1.44 cm^2



2) The formula for the area of a circle is $A = \pi r^2$

Rearrange the formula to make r the subject. Then evaluate to find the radius of a circle with area 80 cm^2.



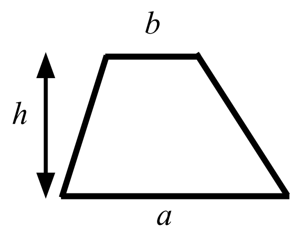
Task B: Evaluating without changing the subject

1) Missing dimensions in a trapezium.

a) A trapezium has area 95 cm^2, a lower base length of 23 cm and an upper base length of 15 cm

Substitute without rearranging to find the height (h).

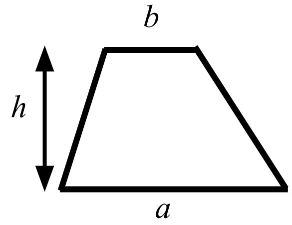
$$A = \frac{1}{2}(a+b)\,h$$



$$A = \frac{1}{2}(a+b)\,h$$

b) A trapezium has area 88 cm^2, a lower base length of 14 cm and a height of 8 cm

Substitute without rearranging to find the upper base (b).

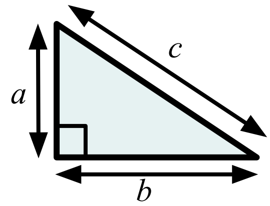


$$A = \frac{1}{2}(a+b)\,h$$

c) Compare the approaches in parts a) and b)

2) A right triangle has hypotenuse 13 cm, and one shorter length of 12 cm

a) Substitute and evaluate without rearranging to find the remaining shorter length.



$$a^2 + b^2 = c^2$$

b) Rearrange to make a the subject, then substitute to find the remaining shorter length.

$$a^2 + b^2 = c^2$$

c) Compare the approaches in parts a) and b)